GY462—Real Estate Finance

Lecture 5

(pre-recorded materials)

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Private Equity Investment Vehicles

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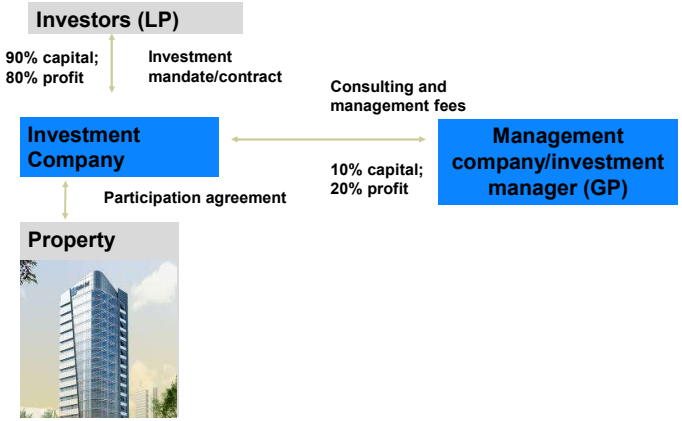
Private equity investment model – *In a nutshell*



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Context and background

- What to do if you want to limit the amount of leverage and/or do not have enough equity?
- Private equity funds: provide equity by raising money from a variety of investors
  - Universities, foundations, pension funds and high net worth individuals
- Funds receive specific investment commitments from investors
  - 'Drawing down' commitments on a 'as needed' basis

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More details

- Investors invest in a limited partnership (LP) managed by the fund sponsor
  - Sponsor acts as general partner (GP) who is responsible for operations and management
- Who are the fund sponsors (GPs)?
  - Investment banks (e.g., UBS); exploit their established contacts and access to deal flows
  - Investment houses (Blackstone, Cerberus…); credibility and reputation are crucial
  - Dedicated RE players (e.g., Lonestar); staff mainly 'veterans' of the sector

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Roles and responsibilities of partners – *Revision*

- LP essentially passive role:
  - Limited involvement in decision making and investments
  - Limited liability (ring-fenced from partnership liability unless involved in management)
- GP controls the partnership:
  - Although it can delegate authority or appoint manager/ advisor to run the fund
  - Unlimited liability – so usually a one-off vehicle (separate GP for each fund to limit cross-exposure)

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Structuring a fund deal – *The intuition*

- When marketing the fund, the sponsor will detail the 'cash waterfall'
  - ✓ That is, how the proceeds from the investment are shared between GP and LP
- Some key terms:
  - Preferred IRR return (or 'pref'): for the investors providing equity (also known as 'the money')
  - 'Promote' or 'Carried interest': goes to the fund mangers

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'Focus point 1'

The workings of private-equity deals:
is the general intuition clear?

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Simple Example – *Linneman (Ch. 17)*

- **Prurhansky Company Pension Fund and Bhavik Fund**
- **'Pref' is 9% going to LP; the split of profits above 9% is 80/20 (investor/manager)**
  - **The (GP) sponsor is part of 'the money'; shares *pari passu* 80% of extra profits**
  - **Two ways of splitting profits; most common: sponsor and 'the money' spilt profits as long as the latter receives all money back + 'pref'**

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Example 1 (Linneman, Ch. 17) – *Continued*

Bhavik Fund Waterfall with 80/20 after 9% Pref is Paid		
Option 1: Split above 9%		
Total Investment		**$ 100 MM**
Prushansky Pension		*$ 90 MM*
Bhavik Fund		*$ 10 MM*
Total Profit		**$ 25 MM**
Preferred Return on Cash: 9%	=100*.9	$ 9 MM
90/100 to Prushansky Pension	=9*.9	*$ 8.10 MM*
10/100 to Bhavik Fund	=9*.1	*$ .90 MM*
Cash Remaining for Split	=25-9	**$ 16 MM**
80% To Money (Prushansky and Bhavik)	=16*.8	*$ 12.80 MM*
90/100 to Prushansky Pension	=12.8*.9	$ 11.52 MM
10/100 to Bhavik Fund	=12.8*.1	$ 1.28 MM
20% To Sponsor (Bhavik)	=16*.2	*$ 3.20 MM*
Total Money to Bhavik Fund		
Money from Pref Return		$ .90 MM
Money from 20% Split		$ 3.20 MM
Money from Pro Rata 80% Split		$ 1.28 MM
		$ 5.38 MM
Total Money to Prushansky Pension		
Money from Pref Return		$ 8.10 MM
Money from Pro Rata 80% Split		$ 11.52 MM
		$ 19.62 MM

Bhavik Fund Waterfall with 80/20 as Long as 9% Paid to Money		
Option 2: "The Money" is above 9%		
Total Investment		**$ 100 MM**
Prushansky Pension		*$ 90 MM*
Bhavik Fund		*$ 10 MM*
Total Profit		**$ 25 MM**
80% To Money (Prushansky and Bhavik)	=25*.8	*$ 20.00 MM*
90/100 to Prushansky Pension	=20*.9	$ 18.00 MM
10/100 to Bhavik Fund	=20*.1	$ 2.00 MM
20% To Sponsor (Bhavik)	=25*.2	*$ 5.00 MM*
Return to Money	=20/100	20%
Total Money to Bhavik Fund		
Money from 20% Split		$ 5.00 MM
Money from Pro Rata 80% Split		$ 2.00 MM
		$ 7.00 MM
Total Money to Prushansky Pension		
Money from Pro Rata 80% Split		**$ 18.00 MM**

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More complex examples with IRR calculation – *Some terminology*

- **Non-cumulative *pari passu* distribution:**
  - **Returns simply flow to GP and LP in proportion to initial investment**
  - **Not very common; mostly 'pref' & 'promote' structures**
  - **Incentivise GP (role of initial GP contribution? Align financial interests)**
- **Cumulative distribution of funds**
  - **If funds insufficient to give investor the preferred yield, liability carries forward to other years**
  - **Investors receive due cash, possibly revalued with interest**

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More complex examples – *Terminology (cont.)*

- **How to share cash from sale?**
  - **First pay debt; then each party reimbursed initial capital**
  - **Then distribute excess in predetermined proportions**
- **IRR preference: cash from sale used to achieve given IRR for some investors**
- **IRR lookback: cash split in proportions if an IRR target is achieved**

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Complex example – *The data*

- **Investor Capital Inc. (ICI) forms joint venture (JV) with Property Devloper Inc. (PDI)**
  - **$50mio investment; ICI: $45 and PDI: $5**
  - **Cash flows as follows**

Year 1	Year 2	Year 3	Year 4	Year 5
$1mio	$2mio	$5mio	$6mio	$6.5mio

  - **Sale in year 5: $75mio**
- **How are revenues shared?**

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Waterfall structure

- **ICI receives a 5% non-cumulative pref**
  - **Non-cum: shortfalls not carried forward**
  - **However, this distribution must be paid before PDI receives anything**
- **After ICI gets pref, PDI receives 5% non-cumulative on its $5**
  - **Extras spilt 50%-50% between ICI and PDI**
- **At sale, ICI recovers capital; then PDI recovers capital**
  - **ICI will also receive enough to achieve 12% IRR; the rest is split 50%-50%**

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Cash flow from operations – *To ICI*

Years	Initial 5% Distribution	50% share in Cash flow – Available after Initial Distribution to ICI and PDI	Total Distribution of Cash Flow from Operations
1	$1,000,000	-- ($5-PDI)* 50%	$1,000,000
2	2,000,000	--	2,000,000
3	2,250,000	$1,250,000	3,500,000
4 $45*5%	2,250,000	1,750,000	4,000,000
5	2,250,000	2,000,000	4,250,000

What if ICI was entitled to receive 'cumulative pref'?

✓ ***Hint: total arrears due in year 3 would be 1,250,000 + 250,000 – Discuss***

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Cash flow from operations – *To PDI and Reconciliation*

To PDI	Initial 5% Distribution	50% Share in Cash flow – Available After Initial Distribution to ICI and PDI	Total Distribution of Cash Flow from Operations
1	--	--	--
2	--	--	--
3	$250,000	$1,250,000	$1,500,000
4	250,000	1,750,000	2,000,000
5	250,000	2,000,000	2,250,000

Reconciliation	Cash for Available for Distribution	Cash Distributed to ICI	Cash Distributed to PDI
1	$1,000,000	$1,000,000	--
2	2,000,000	2,000,000	--
3	5,000,000	3,500,000	$1,500,000
4	6,000,000	4,000,000	2,000,000
5	6,500,000	4,250,000	2,250,000

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Cash flow from sale
– *Achieving 'Target' IRR for ICI*

Years	ICI Income From Operations	ICI Reimbursement of Initial Capital	Additional Cash Flow from Sale for 12% IRR 'Pref' Target	Total Distribution of Cash Flow from Operations
0				-$45,000,000
1	$1,000,000	--	--	$1,000,000
2	2,000,000	--	--	2,000,000
3	3,500,000	--	--	3,500,000
4	4,000,000	--	--	4,000,000
5	4,250,000	$45,000,000	$16,801,668	$66,051,668
				IRR=12%

- **Amount to achieve IRR=12% has been determined; subtracted it from sale price alongside $45mio**
- **The rest is spilt 50-50 between ICI and PDI**

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Final cash flow calculations
– *IRRs for ICI and PDI*

- **Note that ICI and PDI split 50-50 $8,198,332 after 'Pref' 12% IRR is achieved**
- **This is included in year 5 figures**
- **PDI is also reimbursed in year 5 its initial $5mio**

Year	ICI	PDI
0	-$45,000,000	-$5,000,000
1	$1,000,000	0
2	2,000,000	0
3	3,500,000	$1,500,000
4	4,000,000	2,000,000
5	$70,150,834	$11,349,166
IRR	*13.22%*	*26.64%*

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Calculations with lookback
– *IRRs for ICI and PDI*

- **With lookback allocation of sale cash flow is different**
- **Split so that ICI receives overall 12%; rest to PDI**
- **Why should ICI accept a lookback option? Risk & sale value - *Discuss***

Year	ICI	PDI
0	-$45,000,000	-$5,000,000
1	$1,000,000	0
2	2,000,000	0
3	3,500,000	$1,500,000
4	4,000,000	2,000,000
5	$66,051,493	$15,448,507
IRR	*12%*	*32.94%*

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'Focus point 2'

The workings of private-equity deals: are the calculations clear?

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